Supplementary Material for MOSAIC: auditable RNA–ADT annotation with branch-level modality evidence under donor shift and incomplete protein panels

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1 Evidence policy

The primary statistical protocol is nested leave-one-donor-out evaluation. Test donor, validation donor and six training donors are disjoint. A sparse source audit found non-negative, integer-like CSC layers with variable cell totals for both RNA and ADT; the stored layers are therefore treated as raw-count-like inputs without reconstructing upstream download history. The MOSAIC RNA path applies log1p once to the stored layer, selects genes using training donors only and standardizes the selected RNA features with a training-only fit. The locked ADT path applies log1p, a training-only quantile transform to a normal output distribution and a second training-only standardization; the resulting transforms are then applied unchanged to validation and test cells. Source provenance and the complete transform specification are retained in U0. Unless stated otherwise, all intervals are two-sided 95% Student-t intervals across eight held-out donors after averaging three seeds within each held-out donor. Cell counts and seeds are not treated as independent biological observations.

The supplement distinguishes four evidence types: retrained method comparisons, same-checkpoint inference ablations, deterministic missing-panel stress tests and constructed leave-class-out stress tests. PDC101 uses a separate label space and protocol and is reported as a compatible direct hierarchy comparison. Older random-cell, COVID-19, 5P and COMBAT results provide secondary breadth evidence; the donor-disjoint PBMC analysis remains the primary inference.

Supplementary items index. This index makes table numbering, first main-text citation, evidence purpose and source artifacts explicit. The physical order of numbered tables below follows the table numbers.

Item	Short title	First main-text citation	Main use / source fragment or artifact
U0	Preprocessing specification (unnumbered)	Methods, before S1	Transform and training configuration; cache metadata and declared runner arguments
U1	58-label marker/CD8 audit (unnumbered)	Results, rejection and PDC101 paragraphs	Representative panel availability, attribution hits and CD8 boundary linkage; v7 marker audit fragment
S1	Nested splits	Methods, donor-disjoint evaluation	Protocol definition; supplementary source table
S2	Donor performance matrix	Results, donor-disjoint performance	Primary donor metrics; nested-donor performance fragment
S3	No-KD versus matched MLP	Results, component trade-offs	Teacher-dependence audit; no-KD paired audit fragment
S4	Branch-supervision sensitivity	Results, component trade-offs	Branch-loss ablation; branch-supervision fragment
S5	Auditability closure	Results, branch evidence	Conflict, HSR and deletion lift; auditability-closure fragment
S6	Expanded branch-audit scores	Results, branch evidence	Discrete, continuous and combined audit scores; expanded branch-audit fragment
S7	Margin-correctness audit	Results, branch evidence	Branch margin ordering; margin-correctness fragment
S8	Panel masking	Results, panel stress test	Missing-protein stress test; panel-robustness fragment
S9–S11	Leave-class-out rejection	Results, rejection paragraph	Unknown and hierarchy fallback; rejection and decomposition fragments
S12	Light-chain audit	Results, rejection paragraph	B naive lambda failure analysis; light-chain audit fragment
S13–S14	PDC101 comparison	Results, PDC101 paragraph	External hierarchy boundary; PDC101 comparison fragments
S15	Computational cost	Results, evidence map	Runtime and parameter audit; computational-cost fragment
S16–S17	Strong baselines and XGBoost pairing	Results, evidence map	Comparator caveat closure; baseline and paired-comparison fragments
S18	Published latent baselines	Results, evidence map	Fixed-split scANVI and totalVI context; published fixed-split baseline fragment
S19	CLR preprocessing sensitivity	Results, evidence map	ADT preprocessing sensitivity; CLR sensitivity fragment
S20	Attribution stability	Results, evidence map	Marker concordance and faithfulness; attribution fragment
S21	Macro precision/recall decomposition	Results, evidence map	MLP comparison mechanism; macro precision/recall fragment
S22	Protocol-stratified breadth matrix	Results, breadth paragraph	Secondary cross-dataset matrix; declared source matrix
S23	Panel-aware training sensitivity	Supplementary panel-stress section	Training-only ADT feature dropout; panel-aware training fragment
S24	RNA library-size normalization sensitivity	Results, evidence map	Full-transcriptome CP10K/log1p sensitivity; RNA normalization fragment

U0. Preprocessing and primary training specification (unnumbered table).

Component	Locked specification	Leakage control / source
RNA	log1p once; 3,000 genes selected by training-only one-way ANOVA score; training-only StandardScaler	stored layer audit: non-negative integer-like sparse CSC with variable cell totals; upstream download history is not reconstructed
ADT (locked)	log1p; training-only QuantileTransformer to normal output; training-only StandardScaler	transform objects are applied unchanged to validation and test cells
ADT (sensitivity)	per-cell CLR with pseudocount 1.0; training-only StandardScaler	isolated cache; does not replace the locked main row
Model	AdamW, 35-epoch cap, batch 1,024, learning rate 10^{-3}, weight decay 10^{-4}, dropout 0.2, patience 14; constraint-macro selection with ACC/W-F1 tolerance 0.0015; seeds 41/42/43	for each epoch, update the running best ACC and W-F1, score s = val M-F1 $- \rho[\max(0, A^* - A) + \max(0, W^* - W)]$, retain the largest s with deterministic metric tie-breaks, and stop after 14 non-improving epochs; values, seed set and selected checkpoint are declared in each run configuration

2 Dataset lineage and nested splits

The GSE164378 3P and 5P cohorts contain the same eight donors and are not independent-donor datasets. The primary 3P source contains 161,764 cells, 58 L3 labels and 224 proteins. For each fold, 3,000 genes are selected using the six training donors. Every training fold contains all 58 labels, so natural held-out-donor evaluation is closed set.

Table S1: Nested donor split sizes. Validation donors follow a fixed cycle; no donor occurs in more than one partition within a fold. Every training partition contains all 58 L3 labels.

Test donor	Validation donor	Training cells	Validation cells	Test cells	All 58 labels in training
P1	P2	126,424	17,205	18,135	Y
P2	P3	129,899	14,660	17,205	Y
P3	P4	130,014	17,090	14,660	Y
P4	P5	122,863	21,811	17,090	Y
P5	P6	119,211	20,742	21,811	Y
P6	P7	115,151	25,871	20,742	Y
P7	P8	109,643	26,250	25,871	Y
P8	P1	117,379	18,135	26,250	Y

Split manifests are stored under `configs/splits/pbmc_cite_seq/nested_lodo/`. The aggregate cache audit is `results/tables/mosaic_n_v33_nested_lodo_cache_summary_2026-07-23.csv`. Each cache records label support and confirms that feature selection and scalers were fitted on training donors only.

The source audit artifact `results/tables/mosaic_n_v7_pbmc_source_provenance_2026-08-11.csv` records the exact input files and hashes used in the primary run. The RNA layer is 161,764 by 20,729 and the ADT layer is 161,764 by 224; both are sparse CSC float32 matrices with non-negative integer-like entries. The stored per-cell RNA total has minimum/median/maximum 811/6,835/67,793, and the ADT total has 1,001/4,608/49,839. The non-constant totals and integer-like values support treating the stored layers as raw-count-like source matrices; they do not certify the full upstream download history. The locked MOSAIC protocol therefore applies one explicit log1p transform to the stored RNA layer, while the CellTypist compatibility sensitivity separately reconstructs full-transcriptome CP10K before log1p.

3 Complete nested-donor performance matrix

Table S2 reports retrained methods and same-checkpoint inference variants. The point estimate for each donor is the mean of seeds 41, 42 and 43. The 95% interval is then calculated across eight donor means. Worst donor and best donor are the minimum and maximum donor point estimates for performance metrics.

Table S2: Complete nested donor-disjoint performance matrix.

Method	Metric	Mean [95% CI]	Worst donor	Best donor	n donors
ADT branch	ACC	0.8360 [0.8245, 0.8474]	0.8149	0.8550	8
ADT branch	W-F1	0.8364 [0.8270, 0.8457]	0.8159	0.8507	8
ADT branch	M-F1	0.6661 [0.6515, 0.6808]	0.6360	0.6962	8
ADT branch	Balanced accuracy	0.6876 [0.6780, 0.6971]	0.6670	0.6996	8
Fusion branch	ACC	0.9100 [0.9070, 0.9130]	0.9042	0.9163	8
Fusion branch	W-F1	0.9092 [0.9052, 0.9133]	0.9005	0.9157	8
Fusion branch	M-F1	0.8059 [0.7919, 0.8198]	0.7771	0.8242	8
Fusion branch	Balanced accuracy	0.8118 [0.7932, 0.8305]	0.7755	0.8419	8
Learned margin gate	ACC	0.9119 [0.9086, 0.9153]	0.9055	0.9182	8
Learned margin gate	W-F1	0.9107 [0.9062, 0.9153]	0.9010	0.9173	8
Learned margin gate	M-F1	0.8077 [0.7925, 0.8228]	0.7794	0.8314	8
Learned margin gate	Balanced accuracy	0.8096 [0.7906, 0.8286]	0.7756	0.8420	8
MOSAIC full (locked)	ACC	0.9119 [0.9086, 0.9153]	0.9057	0.9183	8
MOSAIC full (locked)	W-F1	0.9108 [0.9062, 0.9153]	0.9012	0.9174	8
MOSAIC full (locked)	M-F1	0.8072 [0.7917, 0.8227]	0.7792	0.8307	8
MOSAIC full (locked)	Balanced accuracy	0.8085 [0.7890, 0.8280]	0.7745	0.8402	8
RNA branch	ACC	0.8894 [0.8846, 0.8942]	0.8827	0.8975	8
RNA branch	W-F1	0.8882 [0.8825, 0.8940]	0.8805	0.8973	8
RNA branch	M-F1	0.7564 [0.7402, 0.7726]	0.7293	0.7782	8
RNA branch	Balanced accuracy	0.7652 [0.7436, 0.7868]	0.7262	0.8062	8
Uniform fusion	ACC	0.9130 [0.9096, 0.9164]	0.9068	0.9196	8
Uniform fusion	W-F1	0.9118 [0.9072, 0.9165]	0.9021	0.9187	8
Uniform fusion	M-F1	0.8114 [0.7961, 0.8266]	0.7803	0.8331	8
Uniform fusion	Balanced accuracy	0.8129 [0.7933, 0.8325]	0.7759	0.8444	8
MLP matched protocol	ACC	0.9103 [0.9065, 0.9141]	0.9036	0.9153	8
MLP matched protocol	W-F1	0.9092 [0.9046, 0.9137]	0.9018	0.9150	8
MLP matched protocol	M-F1	0.7961 [0.7786, 0.8136]	0.7662	0.8240	8
MLP matched protocol	Balanced accuracy	0.8011 [0.7821, 0.8201]	0.7677	0.8297	8
MLP parameter-matched capacity	ACC	0.9111 [0.9081, 0.9142]	0.9059	0.9166	8
MLP parameter-matched capacity	W-F1	0.9096 [0.9051, 0.9142]	0.8999	0.9163	8
MLP parameter-matched capacity	M-F1	0.8001 [0.7846, 0.8156]	0.7721	0.8262	8
MLP parameter-matched capacity	Balanced accuracy	0.8024 [0.7852, 0.8196]	0.7730	0.8321	8
MOSAIC without HSR	ACC	0.9135 [0.9102, 0.9168]	0.9081	0.9200	8
MOSAIC without HSR	W-F1	0.9121 [0.9079, 0.9162]	0.9057	0.9192	8
MOSAIC without HSR	M-F1	0.8097 [0.7950, 0.8243]	0.7847	0.8356	8
MOSAIC without HSR	Balanced accuracy	0.8067 [0.7895, 0.8238]	0.7742	0.8382	8
MOSAIC without KD	ACC	0.9134 [0.9102, 0.9167]	0.9092	0.9200	8
MOSAIC without KD	W-F1	0.9121 [0.9078, 0.9164]	0.9049	0.9194	8
MOSAIC without KD	M-F1	0.8110 [0.7956, 0.8265]	0.7813	0.8360	8
MOSAIC without KD	Balanced accuracy	0.8079 [0.7913, 0.8244]	0.7773	0.8385	8

MOSAIC full versus the matched MLP produced paired differences of +0.0016 ACC $[-0.0011, 0.0044]$ (paired $P = 0.2019$), +0.0016 W-F1 $[-0.0009, 0.0041]$ ($P = 0.1681$), +0.0111 M-F1 $[0.0062, 0.0160]$ ($P = 0.0011$) and +0.0074 balanced accuracy $[0.0013, 0.0135]$ ($P = 0.0239$). Wilcoxon sensitivity P values were 0.3828, 0.3828, 0.0078 and 0.0547, respectively. With eight donor units, the balanced-accuracy gain is treated as marginal because the paired t and Wilcoxon tests differ. Bonferroni adjustment across the four prespecified aggregate endpoints leaves the M-F1 result below 0.005 ($P = 0.0011$); the other endpoints remain non-confirmatory. These tests are not generalized to every exploratory per-class comparison.

The 2,467,706-parameter capacity-control MLP (704/256 hidden dimensions) reached 0.9111/0.9096/0.8001/0.8024 for ACC/W-F1/M-F1/balanced accuracy. MOSAIC minus this control was +0.0008 $[-0.0017, +0.0033]$, +0.0011 $[-0.0011, +0.0033]$, +0.0071 $[+0.0031, +0.0111]$ and +0.0060 $[+0.0001, +0.0120]$, respectively. The M-F1 paired t test was $P = 0.0042$ and the Wilcoxon sensitivity was $P = 0.0156$; this control narrows the capacity confound but does not establish exact parameter-matched architecture causality.

The direct architecture ablations separate aggregate optimization from evidence emission. Full minus no-HSR was −0.0016 ACC, −0.0013 W-F1, −0.0025 M-F1 and +0.0018 balanced accuracy. Full minus no-KD was −0.0015, −0.0014, −0.0038 and +0.0006, respectively. Same-checkpoint learned margin gate plus HSR minus uniform fusion was −0.0011 ACC, −0.0010 W-F1, −0.0042 M-F1 and −0.0044 balanced accuracy. The full paired table is `results/tables/mosaic_n_v50_matched_mlp_paired_2026-08-03.csv`.

Table S3: No-KD versus matched MLP paired donor audit. Differences are MOSAIC without KD minus the matched MLP after averaging seeds within each held-out donor. Balanced accuracy was reconstructed from supported per-class recalls for both methods.

Metric	MOSAIC no KD	MLP matched	Difference [95% CI]	paired P	Wilcoxon P	+ donors
ACC	0.9134	0.9103	+0.0031 [0.0003, 0.0059]	0.03295	0.03906	6/8
W-F1	0.9121	0.9092	+0.0030 [0.0004, 0.0056]	0.02958	0.02344	7/8
M-F1	0.8110	0.7961	+0.0150 [0.0091, 0.0208]	0.0005156	0.007812	8/8
Balanced accuracy	0.8079	0.8011	+0.0068 [0.0003, 0.0133]	0.0434	0.03906	7/8

This comparison separates the family-level gain from the KD term. MOSAIC without KD increased ACC, W-F1, M-F1 and balanced accuracy by +0.0031, +0.0030, +0.0150 and +0.0068, respectively, over the matched MLP; the M-F1 difference was positive on all eight donors, while the other metrics were positive on six, seven and seven donors. In the locked configuration, KD is therefore best understood as part of the evidence-emitting training path rather than the source of the family-level macro gain.

Per-class results exclude donor–class combinations with zero true support rather than assigning F1 zero. The support-aware table is `results/tables/mosaic_n_v33_donor_per_class_summary_2026-07-23.csv`. Predictions into a class absent from a donor's truth set remain penalized in donor-level aggregate metrics.

4 Branch and auditability closure

The branch-supervision sensitivity matrix used the same nested leave-one-donor-out protocol as the primary table. It reused the same train-only caches, validation-donor mapping, seed set and teacher-probability protocol. The tested configurations were $\lambda_b = 0, \lambda_f = 0.5$, $\lambda_b = 0.15, \lambda_f = 0.5$, $\lambda_b = 0.70, \lambda_f = 0.5$ and a fusion-supervision-only setting with $\lambda_b = 0, \lambda_f = 1$. The primary full configuration corresponds to $\lambda_b = 0.35, \lambda_f = 0.5$ and is reported in Table S2.

Table S4: Branch-supervision sensitivity under the primary nested donor protocol. Values are donor means with 95% confidence intervals after averaging three seeds within each donor.

Metric	$\lambda_b = 0, \lambda_f = 0.5$	$\lambda_b = 0.15, \lambda_f = 0.5$	locked $\lambda_b = 0.35, \lambda_f = 0.5$	$\lambda_b = 0.70, \lambda_f = 0.5$	fusion-only $\lambda_b = 0, \lambda_f = 1$
ACC	0.9102 [0.9065, 0.9139]	0.9103 [0.9068, 0.9138]	0.9119 [0.9086, 0.9153]	**0.9140 [0.9111, 0.9169]**	0.9108 [0.9075, 0.9142]
W-F1	0.9095 [0.9052, 0.9137]	0.9094 [0.9053, 0.9135]	0.9108 [0.9062, 0.9153]	**0.9126 [0.9086, 0.9165]**	0.9099 [0.9055, 0.9143]
M-F1	0.8084 [0.7937, 0.8230]	0.8061 [0.7924, 0.8199]	0.8072 [0.7917, 0.8227]	0.8092 [0.7920, 0.8264]	**0.8122 [0.7992, 0.8251]**
Balanced ACC	0.8097 [0.7922, 0.8272]	0.8082 [0.7954, 0.8210]	**0.8085 [0.7890, 0.8280]**	0.8063 [0.7863, 0.8264]	0.8132 [0.7989, 0.8276]

The sensitivity curve identifies a deliberate design trade-off. $\lambda_b = 0.70$ produced the highest ACC and W-F1, while fusion-supervision-only training produced the highest M-F1 and balanced accuracy. The $\lambda_b = 0$ model already retained the dual-path encoder and exceeded the early-fusion MLP by +0.0123 M-F1 and +0.0086 balanced accuracy. Branch supervision adds the modality-specific labels and margins needed for the audit record, with a measured cost of 0.0012 M-F1 and 0.0012 balanced accuracy relative to that $\lambda_b = 0$ configuration; the locked $\lambda_b = 0.35$ configuration preserves this evidence interface.

Table S5: Quantitative auditability closure. Branch-disagreement and uncertainty scores use final prediction error as the target. The pooled held-out error prevalence is 0.0881, and the six declared HSR labels cover 12.35% of held-out cells. HSR action rates are reported both as a pooled cell-level rate (0.14% across all held-out predictions) and as a donor–seed mean (0.15%); attribution faithfulness reports top-feature deletion lift over matched random deletion.

Audit question	Measurement	Donor-level result
Top-label conflict	error AUROC	0.7552 [0.7445, 0.7659]
Top-label conflict	error AUPRC	0.2394 [0.2277, 0.2511]
Top-label conflict	risk at 90% coverage	0.0633 [0.0605, 0.0661]
Final uncertainty	error AUROC	0.6914 [0.6507, 0.7320]
Final uncertainty	error AUPRC	0.3100 [0.2906, 0.3294]
Final uncertainty	risk at 90% coverage	0.0563 [0.0513, 0.0614]
HSR action audit	HSR prediction-change rate	0.0015 [0.0010, 0.0020]
HSR action audit	HSR wrong-to-correct rate	0.0006 [0.0005, 0.0008]
HSR action audit	HSR correct-to-wrong rate	0.0006 [0.0004, 0.0008]
Attribution faithfulness	ADT top-feature deletion lift	0.2358; positive 6/6 tests
Attribution faithfulness	RNA top-feature deletion lift	0.1304; positive 6/6 tests

Top-label conflict improved error-detection AUROC over final uncertainty and provides a direct indicator of modality conflict. Across the 24 full fold–seed artifacts, HSR changed 688 of 485,292 pooled held-out predictions (0.14% overall); the changed fraction was 1.15% within the six declared HSR labels. Wrong-to-correct and correct-to-wrong changes were balanced, showing that the path is sparse and auditable rather than a broad relabeling operation. Same-checkpoint HSR effects were below the prespecified descriptive audit threshold of 0.005 for ACC, W-F1, M-F1 and balanced accuracy. The aggregate and audit roles of HSR are therefore reported separately.

For the audit rows, branch disagreement is the indicator $\mathbb{1}[|\{\hat{y}_r, \hat{y}_p, \hat{y}_f\}| > 1]$ that at least two of the RNA, ADT and fusion branch top labels differ. Final uncertainty is one minus the final maximum softmax probability. Top-probability variance is the variance of the three branch top-class probabilities, and margin variance is the variance of the three branch top-two logit margins. Risk at 90% coverage sorts cells by the corresponding audit score, accepts the 90% lowest-risk cells within each donor–seed unit, and reports the error rate among accepted cells. Ties are resolved by the stable input order and do not affect donor-level means at the reported precision.

Table S6: Expanded error-audit scores. Results are donor-level means with 95% confidence intervals after averaging seeds within each donor. Risk-at-50% rows are pooled operating points from the risk–coverage artifact and are shown without a donor-level interval. Top-label conflict is a discrete function of the three branch top labels; top-probability variance and margin variance are continuous branch-derived scores available from the same cell-level audit artifact. The one- through four-feature audit-subset rows use target-donor-held-out logistic fitting and are descriptive feature ablations. For each target donor, the combined logistic score is fitted on out-of-fold predictions from the seven non-held-out donor folds using final uncertainty, branch conflict, top-probability variance and margin variance, then evaluated on the target held-out donor. The two displayed 0.0565 risk values correspond to different coverage operating points and are a numerical coincidence, not a duplicated estimate.

Score	Type	Measurement	Result	Note
Final uncertainty	continuous	error AUROC	0.6914 [0.6507, 0.7320]	continuous final confidence score
Final uncertainty	continuous	error AUPRC	0.3100 [0.2906, 0.3294]	continuous final confidence score
Final uncertainty	continuous	risk at 90% coverage	0.0563 [0.0513, 0.0614]	continuous final confidence score
Top-label conflict	discrete	error AUROC	0.7552 [0.7445, 0.7659]	discrete top-label conflict score
Top-label conflict	discrete	error AUPRC	0.2394 [0.2277, 0.2511]	discrete top-label conflict score
Top-label conflict	discrete	risk at 90% coverage	0.0637 [0.0605, 0.0669]	discrete top-label conflict score
Top-probability variance	continuous	error AUROC	0.6979 [0.6813, 0.7145]	continuous branch top-probability variance
Top-probability variance	continuous	error AUPRC	0.1466 [0.1341, 0.1590]	continuous branch top-probability variance
Top-probability variance	continuous	risk at 90% coverage	0.0811 [0.0788, 0.0834]	continuous branch top-probability variance
Margin variance	continuous	error AUROC	0.6902 [0.6730, 0.7074]	continuous branch top-two margin variance
Margin variance	continuous	error AUPRC	0.1344 [0.1238, 0.1449]	continuous branch top-two margin variance
Margin variance	continuous	risk at 90% coverage	0.0851 [0.0824, 0.0878]	continuous branch top-two margin variance
Combined logistic score	continuous	error AUROC	0.8401 [0.8267, 0.8536]	logistic combination fitted on out-of-fold predictions from seven non-held-out donor folds
Combined logistic score	continuous	error AUPRC	0.3587 [0.3478, 0.3696]	logistic combination fitted on out-of-fold predictions from seven non-held-out donor folds
Combined logistic score	continuous	risk at 90% coverage	0.0565 [0.0530, 0.0600]	logistic combination fitted on out-of-fold predictions from seven non-held-out donor folds
Audit subset (1 feature)	continuous	error AUROC	0.6915 [0.6514, 0.7317]	LODO logistic score: final uncertainty only
Audit subset (1 feature)	continuous	error AUPRC	0.3080 [0.2885, 0.3275]	LODO logistic score; target donor held out
Audit subset (1 feature)	continuous	risk at 90% coverage	0.0550 [0.0499, 0.0601]	LODO logistic score; target donor held out
Audit subset (2 features)	continuous	error AUROC	0.7860 [0.7685, 0.8036]	LODO logistic score: final uncertainty + branch conflict
Audit subset (2 features)	continuous	error AUPRC	0.3695 [0.3611, 0.3778]	LODO logistic score; target donor held out
Audit subset (2 features)	continuous	risk at 90% coverage	0.0510 [0.0484, 0.0535]	LODO logistic score; target donor held out
Audit subset (3 features)	continuous	error AUROC	0.8379 [0.8249, 0.8509]	LODO logistic score: add top-probability variance
Audit subset (3 features)	continuous	error AUPRC	0.3548 [0.3446, 0.3650]	LODO logistic score; target donor held out
Audit subset (3 features)	continuous	risk at 90% coverage	0.0556 [0.0527, 0.0585]	LODO logistic score; target donor held out
Audit subset (4 features)	continuous	error AUROC	0.8389 [0.8253, 0.8525]	LODO logistic score: add margin variance
Audit subset (4 features)	continuous	error AUPRC	0.3571 [0.3461, 0.3682]	LODO logistic score; target donor held out
Audit subset (4 features)	continuous	risk at 90% coverage	0.0568 [0.0533, 0.0602]	LODO logistic score; target donor held out
Final uncertainty	continuous	risk at 50% coverage	0.0565	pooled risk–coverage curve point; no donor-level interval in source artifact
Combined logistic score	continuous	risk at 50% coverage	0.0169	pooled risk–coverage curve point; no donor-level interval in source artifact
Combined minus final uncertainty	paired	Δ error AUROC	+0.1488 [0.1162, 0.1814]; $P_t = 1.29 \times 10^{-5}$; $P_W = 0.0078$	donor-paired LODO comparison
Combined minus final uncertainty	paired	Δ error AUPRC	+0.0487 [0.0295, 0.0679]; $P_t = 5.47 \times 10^{-4}$; $P_W = 0.0078$	donor-paired LODO comparison
Combined minus final uncertainty	paired	Δ risk at 90% coverage	+0.0002 [-0.0022, 0.0025]; $P_t = 0.871$	donor-paired LODO comparison
All-four logistic	coefficient	final uncertainty	0.634 ± 0.024	mean ± SD across eight LODO fits; standardized coefficient
All-four logistic	coefficient	branch disagreement	0.631 ± 0.015	mean ± SD across eight LODO fits; standardized coefficient
All-four logistic	coefficient	top-probability variance	0.322 ± 0.021	mean ± SD across eight LODO fits; standardized coefficient
All-four logistic	coefficient	margin variance	0.306 ± 0.026	mean ± SD across eight LODO fits; standardized coefficient

The expanded audit shows how the evidence fields complement one another. Top-label conflict is available directly from the three branch top-label votes, while continuous top-probability and margin variance are available from the same cell-level record. The combined logistic score adds these branch fields to final uncertainty. For target donor k, the score was fitted only on cell-level out-of-fold predictions from the seven folds with held-out donors $j \neq k$ and then evaluated on the untouched out-of-fold predictions for donor k. The donor-held-out combined score improves AUROC by $+0.1488$ $[0.1162, 0.1814]$ (paired $P = 1.29 \times 10^{-5}$; Wilcoxon $P = 0.0078$) and AUPRC by $+0.0487$ $[0.0295, 0.0679]$ (paired $P = 5.47 \times 10^{-4}$; Wilcoxon $P = 0.0078$) over final uncertainty, while risk at 90% coverage changes by only $+0.0002$ ($[-0.0022, 0.0025]$, $P = 0.871$). At the pooled 50% operating point, risk was 0.0565 for final uncertainty and 0.0169 for the combined score. A predeclared feature-subset audit found that uncertainty plus branch conflict reached AUROC 0.7860, and adding top-probability variance reached 0.8379; their AUPRC and 90% coverage risk were not monotonic, so the four-feature score is retained as an audit operating point rather than an optimized universal score. Branch evidence is therefore most useful as an additional audit feature combined with confidence.

As a separate model-stability diagnostic, three-seed disagreement was scored as one minus the modal prediction count divided by three and evaluated against majority-vote error, not against the single-checkpoint final label error used above. Across eight donors it reached AUROC 0.6956 $[0.6834, 0.7078]$, AUPRC 0.2494 $[0.2347, 0.2642]$, risk at 50% coverage 0.0500 $[0.0466, 0.0534]$ and risk at 90% coverage 0.0505 $[0.0466, 0.0543]$. This result is kept separate from the evidence score because it measures seed instability and should not be interpreted as an additional branch-audit gain. Standardized coefficients in the four-feature donor-held-out logistic fits were positive for all fields and are reported as eight LODO-fit mean ± SD: final uncertainty 0.634 ± 0.024, branch disagreement 0.631 ± 0.015, top-probability variance 0.322 ± 0.021 and margin variance 0.306 ± 0.026. The coefficient artifact is `results/tables/mosaic_n_v7_audit_feature_ablation_2026-08-12_coefficients.csv`; the feature-subset and seed-disagreement artifacts are `results/tables/mosaic_n_v7_audit_feature_ablation_2026-08-12_summary.csv` and `results/tables/mosaic_n_v7_seed_disagreement_2026-08-12.csv`, respectively.

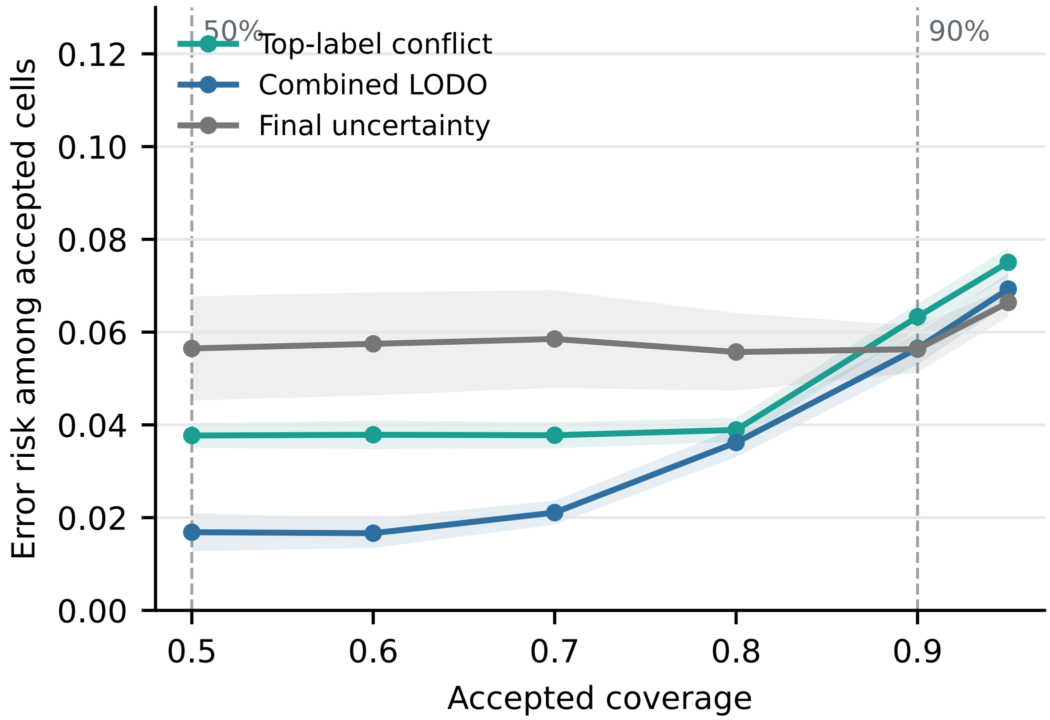

Figure S1: Risk–coverage curves for error auditing. Each curve accepts the lowest-risk cells according to the score on the x-axis coverage fraction and reports the held-out donor error risk among accepted cells. Bands show 95% intervals across eight donors after averaging seeds. The combined LODO score improves the low-coverage region and remains similar to final uncertainty near 90% coverage.

Table S7: Branch margin-correctness audit. Top-two logit margins were computed separately for RNA, ADT and fusion branches in each donor–seed checkpoint. Spearman correlations and quintile summaries use held-out donor as the statistical unit after averaging seeds. Median margins and IQRs are pooled descriptive ranges across donor–seed units; learned β values are the corresponding branch-bias estimates. Monotonic-step intervals are bounded by the four adjacent quintile transitions.

Branch	Accuracy	Spearman	Top–bottom accuracy	Monotonic steps	Median margin (IQR)	β
ADT	0.836 [0.824, 0.847]	0.456 [0.445, 0.468]	0.467 [0.452, 0.481]	3.96 [3.86, 4.00]	2.666 [1.156, 3.730]	-0.501
Fusion	0.910 [0.907, 0.913]	0.289 [0.254, 0.324]	0.273 [0.250, 0.297]	2.46 [1.94, 2.97]	4.467 [3.768, 5.062]	+0.632
RNA	0.889 [0.885, 0.894]	0.388 [0.378, 0.397]	0.372 [0.359, 0.384]	3.62 [3.39, 3.86]	3.496 [2.312, 4.251]	-0.035
Mean over branches	0.878 [0.874, 0.883]	0.378 [0.365, 0.390]	0.370 [0.364, 0.377]	3.35 [3.11, 3.59]	3.543 [2.412, 4.348]	+0.032

The branch margin audit supports the use of branch margins as an ordering signal. All three branches had positive donor-level Spearman correlations between top-two logit margin and branch correctness. The effect was strongest for ADT and RNA and weaker for fusion, consistent with the fusion branch already operating at higher baseline accuracy. Pooled median margins [IQR] were 2.666 [1.156,3.730] for ADT, 3.496 [2.312,4.251] for RNA and 4.467 [3.768,5.062] for fusion; the learned branch biases were −0.501, −0.035 and +0.632, respectively. Margins therefore provide a quantitative bridge from branch prediction to audit prioritization, but remain ordering signals rather than calibrated probabilities.

A gate-bound sensitivity reran the same checkpoints with [0,0.10] and [0,0.50] bounds. Prediction-change rates were 0.034%, 0.131% and 0.149% for [0,0.10], [0,0.50] and the locked [0.02,0.20] range, respectively; paired aggregate differences relative to the locked range remained within the descriptive ±0.005 threshold. This corrected rate counts a changed argmax between base and refined logits, not every non-zero gate value. The summary is in `results/tables/mosaic_n_v7_hsr_gate_bound_summary_2026-08-12.csv` and the paired audit in `results/tables/mosaic_n_v7_hsr_gate_bound_paired_vs_default_2026-08-12.csv`.

The margin-range audit used the saved cell-level top-two logit margins rather than the donor-level bin means. Pooled 5th–95th percentile ranges across the five bins were ADT 0.039–5.930, RNA 0.086–6.940 and fusion 0.205–7.002; the corresponding donor-level Spearman correlations were 0.456, 0.388 and 0.289. The learned branch biases and training-margin means were RNA −0.035/3.782, ADT −0.501/2.697 and fusion +0.632/4.651. These are scale and ordering diagnostics, not calibrated probabilities. The range artifact is `results/tables/mosaic_n_v7_margin_bin_ranges_2026-08-12.csv`.

5 Missing-protein and panel-mismatch stress tests

Twenty-four full checkpoints were evaluated under eight deterministic conditions without retraining or test-label policy selection. Random masks remove the same fixed feature subset for all cells within a fold/seed condition. Marker masks remove six memory-associated or 15 T-cell-associated ADTs after alias normalization.

Table S8: Frozen-checkpoint performance and calibration under protein-panel masks. Intervals use held-out donor after averaging seeds. For ECE and Brier, the worst donor is the maximum value. Marker rows remove six memory-associated or 15 T-cell-associated ADTs; the table explicitly reports removed counts and fractions so targeted and random damage are not conflated. The final two rows are a separate validation-only scalar-temperature audit on the untouched full-input checkpoints: the temperature was fitted by validation negative log likelihood per donor–seed unit and then applied to test probabilities, without changing the argmax predictions.

Scenario	Metric	Removed ADTs	Removed fraction	Mean [95% CI]	ΔM-F1 / fraction	Worst donor	n donors
full	weighted_f1	0	0.0%	0.9108 [0.9062, 0.9153]	–	0.9012	8
full	macro_f1	0	0.0%	0.8072 [0.7917, 0.8227]	0.0000	0.7792	8
full	ece	0	0.0%	0.0713 [0.0645, 0.0781]	–	0.0874	8
full	Brier	0	0.0%	0.1523 [0.1456, 0.1591]	–	0.1629	8
full_val_temperature	ece	0	0.0%	0.0391 [0.0303, 0.0478]	–	0.0527	8
full_val_temperature	Brier	0	0.0%	0.1462 [0.1380, 0.1545]	–	0.1588	8
marker_memory	weighted_f1	6	2.7%	0.9082 [0.9038, 0.9126]	–	0.8991	8
marker_memory	macro_f1	6	2.7%	0.8006 [0.7833, 0.8180]	0.2464	0.7678	8
marker_memory	ece	6	2.7%	0.0738 [0.0670, 0.0806]	–	0.0899	8
marker_memory	Brier	6	2.7%	0.1564 [0.1495, 0.1634]	–	0.1674	8
marker_tcell	weighted_f1	15	6.7%	0.9021 [0.8969, 0.9072]	–	0.8950	8
marker_tcell	macro_f1	15	6.7%	0.7859 [0.7709, 0.8008]	0.3180	0.7647	8
marker_tcell	ece	15	6.7%	0.0748 [0.0664, 0.0832]	–	0.0934	8
marker_tcell	Brier	15	6.7%	0.1661 [0.1591, 0.1731]	–	0.1753	8
random_10	weighted_f1	22	10.0%	0.9074 [0.9015, 0.9132]	–	0.8957	8
random_10	macro_f1	22	10.0%	0.8003 [0.7824, 0.8182]	0.0702	0.7702	8
random_10	ece	22	10.0%	0.0728 [0.0662, 0.0795]	–	0.0892	8
random_10	Brier	22	10.0%	0.1572 [0.1487, 0.1658]	–	0.1704	8
random_30	weighted_f1	67	30.0%	0.9020 [0.8963, 0.9076]	–	0.8920	8
random_30	macro_f1	67	30.0%	0.7904 [0.7742, 0.8066]	0.0562	0.7634	8
random_30	ece	67	30.0%	0.0755 [0.0682, 0.0827]	–	0.0924	8
random_30	Brier	67	30.0%	0.1661 [0.1578, 0.1745]	–	0.1787	8
random_50	weighted_f1	112	50.0%	0.8971 [0.8903, 0.9039]	–	0.8830	8
random_50	macro_f1	112	50.0%	0.7742 [0.7557, 0.7927]	0.0660	0.7424	8
random_50	ece	112	50.0%	0.0764 [0.0669, 0.0859]	–	0.0985	8
random_50	Brier	112	50.0%	0.1741 [0.1647, 0.1835]	–	0.1904	8
random_70	weighted_f1	157	70.0%	0.8921 [0.8855, 0.8988]	–	0.8813	8
random_70	macro_f1	157	70.0%	0.7641 [0.7453, 0.7829]	0.0615	0.7337	8
random_70	ece	157	70.0%	0.0770 [0.0680, 0.0859]	–	0.0965	8
random_70	Brier	157	70.0%	0.1811 [0.1706, 0.1915]	–	0.1954	8
rna_only	weighted_f1	224	100.0%	0.8882 [0.8826, 0.8938]	–	0.8810	8
rna_only	macro_f1	224	100.0%	0.7558 [0.7390, 0.7726]	0.0514	0.7306	8
rna_only	ece	224	100.0%	0.1063 [0.0914, 0.1212]	–	0.1330	8
rna_only	Brier	224	100.0%	0.1888 [0.1826, 0.1950]	–	0.2036	8

The donor-level W-F1 degradation slope over 0/10/30/50/70% random masking was -0.0263 ± 0.0071 per fully missing protein fraction. The M-F1 slope was -0.0624 ± 0.0108. Brier score increased by 0.0411 ± 0.0097 per full missing fraction. ECE slope was less stable ($+0.0081 \pm 0.0072$), whereas the RNA-only ECE increase was clear. Under RNA-only input, the protein vector is zeroed after train-only standardization, so the fused path approaches but is not identical to the RNA branch; residual differences reflect fusion-layer bias terms and learned RNA–ADT interaction weights. The frozen-checkpoint stress test identifies panel-composition sensitivity in the full-input model; a separate training-only panel-aware intervention is evaluated below.

A validation-only scalar-temperature audit was run on the same 24 full checkpoints. One positive temperature was fitted by validation negative log likelihood for each donor–seed unit and then applied to the untouched test probabilities; no test label was used for selection, and the argmax predictions were unchanged in all 24 units. Across held-out donors, test ECE decreased from 0.0713 [0.0645, 0.0781] to 0.0391 [0.0303, 0.0478], NLL from 0.4255 [0.3900, 0.4610] to 0.3904 [0.3432, 0.4376] and Brier score from 0.1523 [0.1456, 0.1591] to 0.1462 [0.1380, 0.1545]. The audit is post-hoc calibration evidence and does not convert the frozen panel-mask stress test into mask-aware training. Unit-level and aggregate files are `results/tables/mosaic_n_v7_temperature_scaling_units_2026-08-12.csv` and `results/tables/mosaic_n_v7_temperature_scaling_2026-08-12.csv`.

To test whether the model can learn a limited panel-mismatch prior, we retrained the same 24 donor–seed units with independent training-only dropout of 30% of standardized ADT features. The feature mask was resampled during minibatch training; evaluation used the complete input or the same deterministic masks as the frozen audit. The donor map, teacher probabilities, loss terms, checkpoint rule and seeds were unchanged, and no test label selected the dropout rate. With the full input restored at evaluation, panel-aware training reached ACC/W-F1/M-F1 0.9098/0.9084/0.8016 versus 0.9119/0.9108/0.8072 for the frozen-checkpoint reference. Under the 30% random mask, it reached 0.9049/0.9034/0.7914 versus 0.9035/0.9020/0.7904; under the targeted T-cell mask, it reached 0.9043/0.9032/0.7884 versus 0.9029/0.9021/0.7859. Thus, the intervention gives a small targeted-mask benefit while imposing a full-input macro trade-off; it is evidence for a bounded panel-aware training sensitivity, not a solution to arbitrary missing-modality deployment. The complete donor-level summary is `results/tables/mosaic_n_v7_panel_aware_training_table_2026-08-12.csv`.

6 Leave-class-out rejection and hierarchy fallback

The five target labels were removed from all training and validation labels before each 57-class model was fitted. Unknown test cells retained the removed label only for evaluation. Thresholds were selected from known validation scores at target known coverage 0.95 or 0.80. No unknown validation or test label informed the threshold.

Table S9: Aggregate validation-selected rejection results. Values first average three seeds within each target and then average five leave-class-out targets from one fixed seed-42 split. The statistical units are the five removed target labels and their seed replicates, not the eight natural PBMC donors. Representative panel-marker availability is a descriptive protocol audit, not a claim that one marker defines the complete cell state.

Score	Target coverage	Observed coverage	Unknown recall	AUROC	Parent-safe rate	Hierarchical accuracy
energy	0.80	0.7961	0.4024	0.6066	0.2076	0.9073
energy	0.95	0.9426	0.2881	0.6066	0.1369	0.8967
one_minus_margin	0.80	0.7878	0.4032	0.5948	0.2259	0.9146
one_minus_margin	0.95	0.9409	0.2139	0.5948	0.1140	0.8994
one_minus_max_probability	0.80	0.7887	0.3643	0.5848	0.1981	0.9101
one_minus_max_probability	0.95	0.9414	0.2372	0.5848	0.1231	0.8990

Table S10: Per-target margin-score results at target known coverage 0.80. Support and target difficulty vary substantially; these five target labels are the stress-test units. The final column records whether a representative panel marker for the target boundary is available; it is a descriptive panel audit rather than a complete marker definition.

Target	n unknown	Known coverage	Unknown recall	AUROC	Parent-safe rate	Hierarchical accuracy	Panel marker
B naive lambda	83	0.7927	0.1285	0.4258	0.0723	0.9277	N: kappa/lambda
CD4 TCM_1	734	0.7807	0.5722	0.7284	0.2489	0.9167	Y: CD4/RO/27
CD8 TEM_4	1291	0.7897	0.4056	0.5696	0.3945	0.9011	Y: CD8/RO/27
NK_3	614	0.7847	0.3458	0.5474	0.3371	0.9192	Y: CD56/335
gdT_2	583	0.7914	0.5638	0.7026	0.0766	0.9083	Y: TCR-V

B naive lambda is the smallest unknown set ($n = 83$) and produced below-chance AUROC and low recall. CD4 TCM_1 and gdT_2 were easier to rank as unknown, but gdT_2 had low parent-safe fallback. The mean unknown parent-safe rate at the chosen margin operating point was 0.2259. This target dependence identifies marker availability as a boundary condition for rejection and motivates manual review for uncertain outputs.

Combined hierarchical accuracy is a stress-test summary alongside closed-set L3 accuracy. Accepted known cells are scored by L3 correctness, while rejected cells from the removed target are scored by parent correctness. Known coverage, unknown recall, AUROC and parent-safe rate are reported alongside the combined score so that parent fallback remains tied to the underlying rejection behavior.

Table S11: Decomposition of the default margin-based leave-class-out operating point. The top block decomposes the mean default margin@0.80 row across five target labels and three seeds per target. The target block reports coverage / unknown recall / AUROC / parent-safe rate / combined hierarchical accuracy so that the combined score cannot be read without target support and unknown-specific behavior.

Default margin@0.80 measurement	Mean	95% CI
Observed known coverage	0.7878	[0.7816, 0.7941]
Unknown recall	0.4032	[0.1767, 0.6297]
Unknown AUROC	0.5948	[0.4415, 0.7480]
Accepted-known L3 accuracy	0.9496	[0.9410, 0.9581]
Accepted-known L3 risk	0.0504	[0.0419, 0.0590]
Rejected-known parent accuracy	0.8936	[0.8813, 0.9058]
Rejected-unknown parent-safe rate	0.2259	[0.0425, 0.4092]
Rejected-and-parent-safe unknown fraction	0.0911	–
Combined hierarchical accuracy	0.9146	[0.9019, 0.9273]
Removed target	N unknown	Coverage / recall / AUROC / parent-safe / combined
B naive lambda	83	0.7927 / 0.1285 / 0.4258 / 0.0723 / 0.9277
CD4 TCM_1	734	0.7807 / 0.5722 / 0.7284 / 0.2489 / 0.9167
CD8 TEM_4	1291	0.7897 / 0.4056 / 0.5696 / 0.3945 / 0.9011
NK_3	614	0.7847 / 0.3458 / 0.5474 / 0.3371 / 0.9192
gdT_2	583	0.7914 / 0.5638 / 0.7026 / 0.0766 / 0.9083

Table S12: Light-chain feature availability for interpreting the B naive lambda leave-class-out failure. RNA markers are checked after each train-only 3,000-gene selection. ADT reports whether a kappa/lambda-like surface marker is present in the 224-protein panel.

Feature	Type	Present donors
kappa_or_lambda_surface_marker	ADT	0/8
IGKC	RNA	8/8
IGLC1	RNA	0/8
IGLC2	RNA	8/8
IGLC3	RNA	8/8
IGLC4	RNA	0/8
IGLC5	RNA	0/8
IGLC6	RNA	0/8
IGLC7	RNA	0/8

IGKC, IGLC2 and IGLC3 were retained in every donor fold, whereas IGLC1 and IGLC4–7 were not retained. The ADT panel did not contain a kappa/lambda surface marker. This supports a concrete interpretation for the B naive lambda pseudo-unknown failure: the class boundary is light-chain specific, and the multimodal input lacks a directly matched protein marker.

58-label representative panel-evidence and CD8 boundary audit

The marker audit below was generated from all 24 current MOSAIC full checkpoints (eight held-out donors and seeds 41/42/43). For each PBMC L3 label and modality, it records a prespecified representative marker-family list, panel availability, top-20 gradient-times-input attribution hits among up to eight correctly predicted cells per class and checkpoint, donor-level M-F1 and mean fusion weight. The representative map is an audit vocabulary rather than a complete marker ontology; it does not define a class or establish causal biology. The five leave-class-out target rows are joined only descriptively to unknown recall and AUROC. The CD8 rows align PBMC labels with the separately evaluated PDC101 classes to expose a boundary-level comparison; the label spaces, preprocessing and uncertainty estimators are not identical.

58-label representative panel-evidence audit (unnumbered diagnostic). The first table covers every current PBMC L3 label and reports the number of representative markers present in the corresponding 3,000-gene or 224-ADT feature space, the number of those markers among the top-20 current-checkpoint attribution features, the donor-level M-F1 and mean fusion weight. The marker lists are representative family tokens, not a complete ontology; availability is not evidence of causal biology.

Label	Group	Avail.	RNA hits	ADT hits	M-F1	Fusion w
ASDC_mDC	none	–	0	0	0.7667	0.631
ASDC_pDC	none	–	0	0	0.7905	0.603
B intermediate kappa	b_cell	5/5	0	1	0.8490	0.764
B intermediate lambda	b_cell	5/5	0	1	0.8281	0.761
B memory kappa	b_cell	7/7	0	2	0.9018	0.736
B memory lambda	b_cell	7/7	0	2	0.8697	0.744
B naive kappa	b_cell	7/7	0	4	0.8999	0.750
B naive lambda	b_cell	7/7	0	3	0.8508	0.743
CD14 Mono	mono	5/5	3	3	0.9859	0.657
CD16 Mono	mono	5/5	0	2	0.9415	0.678
CD4 CTL	cd4	2/2	1	2	0.8677	0.782
CD4 Naive	cd4	5/5	1	5	0.9673	0.695
CD4 Proliferating	none	–	0	0	0.7648	0.614
CD4 TCM_1	cd4	5/5	1	5	0.8303	0.717
CD4 TCM_2	cd4	5/5	1	4	0.7274	0.666
CD4 TCM_3	cd4	5/5	2	4	0.8571	0.754
CD4 TEM_1	cd4	5/5	1	3	0.7761	0.735
CD4 TEM_2	cd4	5/5	1	2	0.6921	0.771
CD4 TEM_3	cd4	5/5	1	2	0.7484	0.734
CD4 TEM_4	cd4	5/5	0	1	0.3715	0.579
CD8 Naive	cd8	5/5	2	5	0.9795	0.714
CD8 Naive_2	cd8	5/5	2	4	0.6305	0.721
CD8 Proliferating	none	–	0	0	0.6579	0.660
CD8 TCM_1	cd8	5/5	2	3	0.6932	0.717
CD8 TCM_2	cd8	5/5	2	2	0.8175	0.747
CD8 TCM_3	cd8	5/5	2	3	0.8449	0.726
CD8 TEM_1	cd8	5/5	2	3	0.8752	0.744
CD8 TEM_2	cd8	5/5	2	3	0.7437	0.733
CD8 TEM_3	cd8	5/5	1	1	0.6086	0.708
CD8 TEM_4	cd8	5/5	2	3	0.7827	0.746
CD8 TEM_5	cd8	5/5	2	3	0.5913	0.724
CD8 TEM_6	cd8	5/5	2	2	0.8578	0.744
Doublet	none	–	0	0	0.6444	0.642
Eryth	eryth	2/2	3	2	0.8482	0.620
HSPC	hspc	3/3	1	3	0.9805	0.705
ILC	ilc	2/2	2	1	0.9419	0.798
MAIT	mait	4/4	1	3	0.9587	0.738
NK Proliferating	none	–	0	0	0.8857	0.722
NK_1	nk	4/4	0	2	0.6560	0.748
NK_2	nk	4/4	0	2	0.9087	0.710
NK_3	nk	4/4	0	3	0.6059	0.714
NK_4	nk	4/4	0	4	0.6993	0.721
NK_CD56bright	nk	4/4	0	3	0.8827	0.755
Plasma	plasma	3/3	2	1	0.9808	0.660
Plasmablast	plasma	3/3	0	1	0.9680	0.594
Platelet	platelet	3/3	2	3	0.9383	0.635
Treg Memory	treg	5/5	0	3	0.8162	0.725
Treg Naive	treg	7/7	2	5	0.7652	0.746
cDC1	dc	3/3	0	1	0.9774	0.729
cDC2_1	dc	3/3	2	2	0.7038	0.744
cDC2_2	dc	3/3	2	2	0.9315	0.687
dnT_1	none	–	0	0	0.4957	0.645
dnT_2	none	–	0	0	0.9268	0.796
gdT_1	gd_t	3/3	3	3	0.9558	0.763
gdT_2	gd_t	3/3	3	1	0.4083	0.725
gdT_3	gd_t	3/3	2	1	0.9208	0.780
gdT_4	gd_t	3/3	2	1	0.7205	0.716
pDC	pdc	4/4	0	3	0.9952	0.663

Target	Panel representative marker	M-F1	Unknown recall	AUROC
B naive lambda	N	0.8508	0.1285	0.4258
CD4 TCM_1	Y	0.8303	0.5722	0.7284
CD8 TEM_4	Y	0.7827	0.4056	0.5696
NK_3	Y	0.6059	0.3458	0.5474
gdT_2	Y	0.4083	0.5638	0.7026

The five-target table is a descriptive stress test rather than a powered association test. B naive lambda is the only target without a direct kappa/lambda-like ADT marker and also has the lowest unknown recall; the target count is too small to infer a general marker–rejection law.

PBMC label	PDC101 class	ADT avail./rep.	ADT hits	Fusion w	MOSAIC F1	MMoCHi F1	Δ F1
CD8 Naive	cd8_n	5/5	5	0.714	0.9260	0.9734	0.0475
CD8 TCM_1	cd8_cm	5/5	3	0.717	0.8246	0.8851	0.0605
CD8 TEM_4	cd8_em	5/5	3	0.746	0.8825	0.9122	0.0297

The CD8 rows connect current PBMC attribution and fusion summaries to the separately evaluated PDC101 hierarchy. MMoCHi's larger F1 in the matched CD8 memory rows is compatible with its curated hierarchy and marker regime, but the label spaces and preprocessing are not identical; this is a boundary comparison, not a causal mediation analysis.

7 PDC101 comparison with official MMoCHi

PDC101 preprocessing was audited before the final comparison. The valid MOSAIC runner uses the processed matrices, fits only StandardScaler on training cells, excludes HTO/isotype/control proteins and validates every official external-holdout identifier and truth label. The MMoCHi row uses the official hierarchy workflow and its native preprocessing path, including its landmark-registration workflow where applicable (MMoCHi 0.3.5, commit `b0f3387`); MOSAIC uses the supplied processed ADT matrix plus train-only standardization. Table S14 is therefore a same-holdout compatibility comparison with package-specific transforms, and the per-class pattern is interpreted as a difference between operating regimes rather than a model-structure-only effect.

Table S14: PDC101 same-holdout aggregate comparison. Uncertainty modes are not interchangeable.

Method	Metric	Mean [95% CI]	Mode
MOSAIC	accuracy	0.9077 [0.8897, 0.9257]	seed Student-t
MOSAIC	weighted_f1	0.9068 [0.8881, 0.9255]	seed Student-t
MOSAIC	macro_f1	0.9007 [0.8798, 0.9216]	seed Student-t
MMoCHi	accuracy	0.9347 [0.9237, 0.9457]	cell bootstrap; conditional on one workflow
MMoCHi	weighted_f1	0.9335 [0.9221, 0.9449]	cell bootstrap; conditional on one workflow
MMoCHi	macro_f1	0.9298 [0.9180, 0.9416]	cell bootstrap; conditional on one workflow

Table S15: PDC101 per-class F1 point estimates on the 2,098-cell sorted holdout. The final column reports the per-class MOSAIC–MMoCHi F1 difference and is ordered from the largest negative gap to the tied class.

Class	Support	MOSAIC F1	MMoCHi F1	Δ F1 (MOSAIC–MMoCHi)
cd8_emra	206	0.8863	0.9493	-0.0630
cd8_cm	223	0.8246	0.8851	-0.0605
cd8_n	241	0.9260	0.9734	-0.0474
cd8_em	365	0.8825	0.9122	-0.0297
cd4_cm	207	0.8185	0.8435	-0.0250
cd4_em	295	0.9374	0.9420	-0.0046
cd4_n	227	0.9333	0.9358	-0.0025
monocyte	334	0.9970	0.9970	+0.0000

MMoCHi exceeded MOSAIC for every aggregate metric and most classes. The largest gaps occurred at CD8 central-memory/effector-memory boundaries, while monocyte F1 was identical. This pattern complements the PBMC audit: curated marker hierarchies are especially effective when the sorted label space and marker program are tightly aligned. MMoCHi intervals are cell bootstrap intervals conditional on one official workflow fit; MOSAIC intervals are Student-t margins over three model seeds.

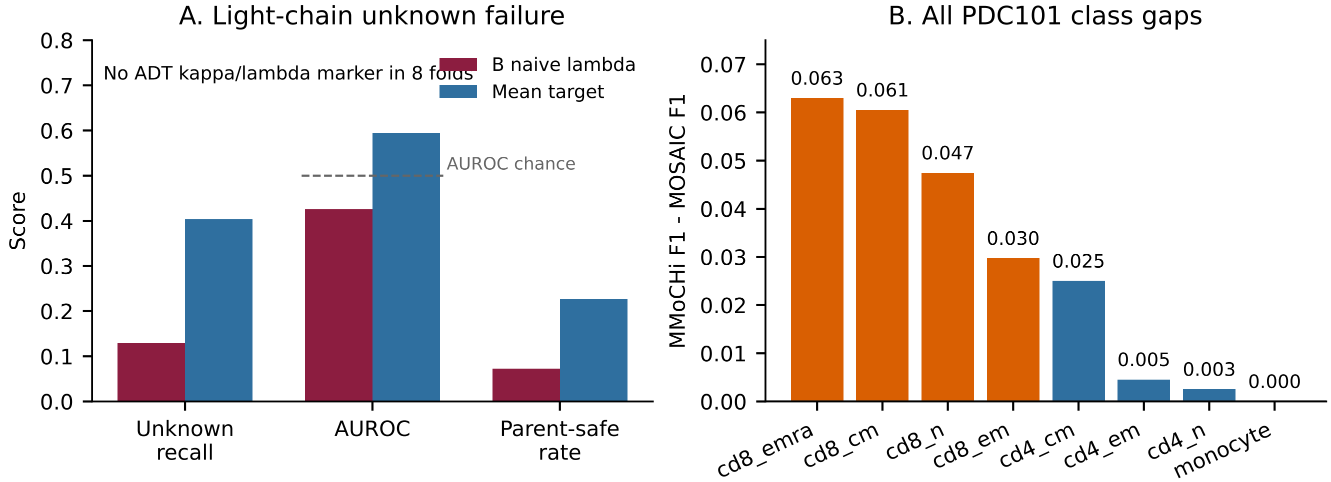

Figure S2: Case-study evidence for marker-limited behavior. (a) B naive lambda was the hardest leave-class-out target, with low unknown recall, below-chance AUROC and low parent-safe rate; the ADT panel contained no kappa/lambda-like marker in any donor fold. (b) On PDC101, the largest MMoCHi–MOSAIC F1 gaps were concentrated in CD8 memory/effector-memory-related classes, whereas monocyte F1 was identical.

These two cases show how evidence availability shapes model behavior. B naive lambda illustrates absent marker evidence for unknown detection, whereas the PDC101 CD8 boundary illustrates a setting where an official curated hierarchy better matches the label structure than a learned high-dimensional reference classifier.

8 Computational cost

Runtime summaries were rebuilt from the primary nested-donor `results_summary.csv` artifacts. Each run corresponds to one held-out donor and one seed. Parameter counts use the primary configuration dimensions: 3,000 RNA genes, 224 ADTs, 58 classes, MLP hidden dimensions 512/128, the capacity-control MLP hidden dimensions 704/256, and MOSAIC hidden dimension 256 with branch encoders, fusion encoder and linear L3 heads. Runtime does not include raw-data download or public release packaging. The no-HSR row has fewer parameters than MOSAIC full but a later median checkpoint epoch (30.0 versus 25.0), which explains why its median runtime is longer despite the smaller parameter count.

Table S16: Computational cost in the primary nested donor matrix. Runtime is wall-clock training time per fold–seed unit recorded in the run summaries.

Method	Parameters	Runs	Median runtime (s)	Runtime range (s)	Median epoch	Device
MLP matched protocol	1.73M	24	75.8	50.6–86.9	15.5	cuda
MLP parameter-matched capacity	2.47M	24	80.7	51.6–89.5	20.5	cuda
MOSAIC full	2.51M	24	130.0	89.5–201.1	25.0	cuda
MOSAIC without HSR	2.42M	24	146.0	82.6–179.4	30.0	cuda
MOSAIC without KD	2.51M	24	135.9	89.7–182.3	25.5	cuda

The locked MOSAIC full path was benchmarked on the P1/seed-42 checkpoint using a 4,096-cell GPU batch. Prediction-only throughput was 3.34×10^6 cells/s, while materializing the branch predictions, top-two margins, weights, confidence fields, HSR gate and delta norm for the audit record reached 2.97×10^6 cells/s, corresponding to a 12.6% wall-time overhead. This microbenchmark measures inference and record extraction only, not training or data loading; the artifact is `results/tables/mosaic_n_v7_audit_throughput_2026-08-12.csv`.

Inference-only throughput and audit-record extraction overhead.

Mode	Cells per batch	Cells/s	Audit overhead
Prediction only	4096	3.34×10^6	–
Prediction + audit record	4096	2.97×10^6	12.6%

9 Full-training published and strong ML donor baselines

To broaden the comparator matrix, we reran a CellTypist-compatible RNA-only comparator and nine classical or tree-based baselines on the primary nested donor protocol using all cells from the six training donors in each fold. Each method was evaluated on the complete held-out donor, and no validation or test label was used for model selection. The primary CellTypist-compatible row applies CP10K using the full-transcriptome RNA denominator, log1p transforms the result and then retains the train-only selected 3,000 genes without gene-wise z-scoring. The runner calls the CellTypist `train/annotate` API with $C = 1.0$, `max_iter=20`, `use_SGD=True`, `alpha=1e-4`, `n_jobs=8`, `balance_cell_type=True`, `feature_selection=False`, no majority voting and the expression-sum check disabled only because the post-selection matrix no longer sums to 10,000 counts; this is a package-compatible implementation rather than an unqualified native-default claim. The subset-first convention is retained as a preprocessing sensitivity: raw counts are first restricted to the selected genes and only then CP10K-normalized. XGBoost, ridge and nearest-centroid variants were run separately for RNA-only, ADT-only and early-fusion RNA+ADT inputs. The corrected XGBoost runs remove the former 500-cells-per-class cap and use 80 trees, depth 4, learning rate 0.08, subsample 0.9, column subsample 0.8, histogram training with CUDA and eight workers; prediction/evaluation consumes the materialized CPU arrays. No XGBoost hyperparameter was selected from held-out labels; the configuration was fixed before donor evaluation. The exact input conventions are retained in the table caption and run artifacts.

R/Bioconductor scmap and scPred were not used as primary numeric comparators in this table because their current artifacts are protocol-specific tool-coverage records rather than complete uncapped donor matrices.

Table S17: Full-training published and strong ML baselines under the primary PBMC donor-disjoint L3 protocol. Values are donor means with 95% confidence intervals across eight held-out donors. All listed baselines use the full six training donors in each fold and the complete held-out donor test set. The XGBoost rows are uncapped and use the declared 80-tree, depth-4 histogram configuration. The primary CellTypist-compatible row uses full-transcriptome-denominator CP10K followed by log1p before retaining the train-only 3,000-gene subset; the subset-first row is a sensitivity and neither row is presented as an unqualified fully native package-default reproduction.

Method	Modality	ACC	W-F1	M-F1	B-ACC	Worst ACC	Best ACC
CellTypist-compatible subset-first	RNA-only	0.8652 [0.8573, 0.8732]	0.8565 [0.8477, 0.8654]	0.6559 [0.6249, 0.6870]	0.6240 [0.5950, 0.6530]	0.8486	0.8767
CellTypist-compatible full-denominator	RNA-only	0.8534 [0.8391, 0.8677]	0.8497 [0.8350, 0.8644]	0.6835 [0.6608, 0.7062]	0.6933 [0.6731, 0.7136]	0.8310	0.8712
Early-fusion XGBoost (uncapped)	RNA+ADT	0.9032 [0.8912, 0.9152]	0.9010 [0.8901, 0.9120]	0.7885 [0.7648, 0.8122]	0.7755 [0.7543, 0.7968]	0.8750	0.9201
RNA XGBoost (uncapped)	RNA-only	0.8741 [0.8637, 0.8846]	0.8681 [0.8566, 0.8796]	0.7168 [0.6969, 0.7368]	0.6967 [0.6743, 0.7192]	0.8474	0.8979
ADT XGBoost (uncapped)	ADT-only	0.8426 [0.8313, 0.8540]	0.8349 [0.8254, 0.8444]	0.6557 [0.6371, 0.6742]	0.6429 [0.6273, 0.6584]	0.8185	0.8679
Early-fusion ridge	RNA+ADT	0.8547 [0.8442, 0.8652]	0.8571 [0.8471, 0.8672]	0.7028 [0.6814, 0.7242]	0.7660 [0.7556, 0.7763]	0.8349	0.8720
RNA ridge	RNA-only	0.8329 [0.8179, 0.8479]	0.8373 [0.8245, 0.8500]	0.6706 [0.6486, 0.6925]	0.7412 [0.7303, 0.7520]	0.8067	0.8587
ADT ridge	ADT-only	0.7595 [0.7411, 0.7778]	0.7626 [0.7448, 0.7803]	0.5072 [0.4819, 0.5326]	0.6460 [0.6280, 0.6640]	0.7299	0.7952
Early-fusion centroid	RNA+ADT	0.7244 [0.6581, 0.7907]	0.7395 [0.6826, 0.7963]	0.5787 [0.5188, 0.6386]	0.6269 [0.5688, 0.6851]	0.6260	0.8373
RNA centroid	RNA-only	0.7280 [0.6612, 0.7948]	0.7509 [0.6991, 0.8026]	0.5809 [0.5293, 0.6324]	0.6207 [0.5685, 0.6729]	0.6282	0.8231
ADT centroid	ADT-only	0.3761 [0.2661, 0.4862]	0.4219 [0.3234, 0.5204]	0.2669 [0.2026, 0.3312]	0.3433 [0.2679, 0.4188]	0.1858	0.5267

The full-transcriptome CP10K/log1p CellTypist run reached ACC/W-F1/M-F1/balanced accuracy 0.8534/0.8497/0.6835/0.6933. The subset-first sensitivity reached 0.8652/0.8565/0.6559/0.6240, showing that normalization order changes the comparator profile. Uncapped early-fusion XGBoost reached 0.9032/0.9010/0.7885/0.7755 for the same metrics. Relative to it, MOSAIC's ACC difference was $+0.0087$ $[-0.0029, +0.0204]$, with corresponding differences of $+0.0097$ W-F1, $+0.0187$ M-F1 and $+0.0330$ balanced accuracy; exact intervals, Cohen d_z, paired t and Wilcoxon values are in Table S18. The old capped row is not retained as a main result.

Table S18: Paired donor comparison between MOSAIC full and the strongest full-training baseline, early-fusion XGBoost. Differences are MOSAIC minus XGBoost after averaging MOSAIC seeds within each held-out donor.

Metric	Paired difference [95% CI]	Cohen d_z	paired t P	Wilcoxon P	+ donors
ACC	0.0087 [-0.0029, 0.0204]	0.629	0.1183	0.1094	6/8
W-F1	0.0097 [-0.0010, 0.0205]	0.755	0.07016	0.05469	6/8
M-F1	0.0187 [0.0053, 0.0321]	1.167	0.01311	0.01562	7/8
B-ACC	0.0330 [0.0190, 0.0469]	1.977	0.0008238	0.007812	8/8

MOSAIC full had positive point estimates relative to early-fusion XGBoost for ACC, W-F1 and M-F1; the paired donor intervals quantify the remaining uncertainty around this point-estimate advantage.

10 Published latent baselines under a fixed strict split

To add published representation-learning context, we completed auditable scANVI and totalVI runs on the fixed seed-42 strict PBMC split. These runs use 15,000 held-out test cells; scANVI uses 1,000 RNA genes, whereas totalVI uses 1,000 RNA genes and 224 proteins followed by a latent-space logistic readout. Their fixed-split results complement the nested donor matrix and are reported with their split and readout metadata.

Table S19: Protocol-caveated published baselines on a fixed strict PBMC split. Values are test-set metrics from one seed-42 run; the split is not nested donor-LODO and is not pooled with the primary donor-level inference.

Method	Modality	Split	Readout	ACC	W-F1	M-F1	B-ACC
scANVI	RNA	fixed seed 42	latent classifier	0.8907	0.8882	0.7729	0.7502
totalVI	RNA+ADT	fixed seed 42	latent logistic	0.8100	0.8251	0.6916	0.7699

The fixed-split scANVI result was ACC 0.8907, W-F1 0.8882, M-F1 0.7729 and balanced accuracy 0.7502. totalVI reached ACC 0.8100, W-F1 0.8251, M-F1 0.6916 and balanced accuracy 0.7699 under its latent logistic readout. These values provide protocol-specific representation-learning context and are not pooled with the donor-averaged primary estimates.

The official MMoCHi PBMC audit was also completed as a label-space check. MMoCHi's official hierarchy returns broad lineage nodes rather than the 58 PBMC L3 labels: 44 labels can only be mapped to broad subsets and 14 are not covered, with zero direct official L3 outputs. We therefore do not impute a PBMC L3 metric or call a label-derived adaptation an official MMoCHi baseline. The auditable disposition is recorded in `results/tables/mmochi_pbmc_strict_adaptation_audit_2026-05-18.csv`.

ADT preprocessing sensitivity We repeated the full eight-fold, three-seed nested donor matrix with a separate CLR cache. For each cell i and protein j, protein values were transformed as $\mathrm{CLR}(x_{ij}) = \log(x_{ij}+1) - \frac{1}{P}\sum_{j'=1}^{P}\log(x_{ij'}+1)$, followed by a training-donor-only standardization; no test-cell statistics entered the transform. The locked quantile-normalized protocol and the CLR protocol therefore differ only in the ADT transform, while donor mapping, seeds, train/validation reference, model configuration and checkpoint selection were retained. The compact summary below reports donor means and two-sided 95% Student-t intervals.

Table S20: Locked versus CLR ADT preprocessing sensitivity. Each method has 24 fold–seed units (eight held-out donors and three seeds); values are donor means with two-sided 95% Student-t intervals. The paired MOSAIC–MLP comparison is reported in the text below.

Protocol	Method	ACC	W-F1	M-F1	B-ACC
locked	MOSAIC full locked	0.9119 [0.9086, 0.9153]	0.9108 [0.9062, 0.9153]	0.8072 [0.7917, 0.8227]	0.8085 [0.7890, 0.8280]
clr	MOSAIC full CLR	0.9150 [0.9120, 0.9180]	0.9136 [0.9094, 0.9179]	0.8176 [0.7987, 0.8365]	0.8200 [0.8022, 0.8379]
locked	MLP matched locked	0.9103 [0.9065, 0.9141]	0.9092 [0.9046, 0.9137]	0.7961 [0.7786, 0.8136]	0.8011 [0.7821, 0.8201]
clr	MLP matched CLR	0.9134 [0.9097, 0.9172]	0.9127 [0.9083, 0.9170]	0.8112 [0.7974, 0.8250]	0.8175 [0.8036, 0.8315]
CLR paired MOSAIC–MLP	difference	+0.0016 [-0.0006, 0.0038]	+0.0009 [-0.0016, 0.0035]	+0.0064 [-0.0012, 0.0139]	+0.0025 [-0.0039, 0.0089]

CLR increased the locked MOSAIC row by 0.0031 ACC, 0.0029 W-F1, 0.0104 M-F1 and 0.0116 balanced accuracy; the corresponding matched-MLP changes were 0.0031, 0.0035, 0.0151 and 0.0164. Under CLR, MOSAIC minus matched MLP was +0.0016 ACC [−0.0006, +0.0038], +0.0009 W-F1 [−0.0016, +0.0035], +0.0064 M-F1 [−0.0012, +0.0139] (paired $P = 0.0861$) and +0.0025 balanced accuracy [−0.0039, +0.0089]; all four paired intervals crossed zero. CLR improved both methods in absolute M-F1, with a larger increase for the MLP. This sensitivity analysis establishes preprocessing as a first-class source of variation in both absolute performance and the comparator gap; the locked transform remains the prespecified primary protocol. DSB requires matched background/isotype controls not available in this dataset.

11 Cross-donor feature attribution

Gradient-times-input was aggregated separately for RNA and ADT in six prespecified sibling labels. Donor-pair values first average three seeds. Because the 28 donor pairs overlap, Spearman and Jaccard summaries are descriptive and are not used as 28 independent replicates.

Table S21: Cross-donor attribution stability and canonical-marker enrichment. Marker overlap is the fraction of available canonical markers appearing in the aggregate top 20. BH-FDR values adjust the 12 marker-enrichment permutation tests.

Class	Modality	Mean Spearman	Top-20 Jaccard	Marker overlap	Permutation p	BH-FDR q
CD4 TCM_3	ADT	0.6123	0.5103	1.0000	2.0×10^{-4}	6.0×10^{-4}
CD4 TCM_3	RNA	0.7510	0.4355	0.4000	0.0012	0.0024
CD4 TEM_3	ADT	0.5535	0.4722	0.5000	0.0476	0.0571
CD4 TEM_3	RNA	0.7400	0.4182	0.5000	2.0×10^{-4}	6.0×10^{-4}
CD4 TEM_4	ADT	0.2945	0.2915	0.2500	0.3181	0.3471
CD4 TEM_4	RNA	0.5292	0.1127	0.5000	4.0×10^{-4}	9.6×10^{-4}
CD8 Naive	ADT	0.5545	0.4684	0.8000	2.0×10^{-4}	6.0×10^{-4}
CD8 Naive	RNA	0.7956	0.2895	0.0000	1.0000	1.0000
CD8 Naive_2	ADT	0.4850	0.4740	0.8000	2.0×10^{-4}	6.0×10^{-4}
CD8 Naive_2	RNA	0.6829	0.2993	0.2000	0.0300	0.0408
CD8 TCM_1	ADT	0.5816	0.4810	0.6000	0.0064	0.0110
CD8 TCM_1	RNA	0.7529	0.4548	0.2000	0.0306	0.0408

RNA rank stability was highest for CD8 Naive (mean Spearman 0.796) and lowest for CD4 TEM_4 (0.529). ADT stability ranged from 0.295 for CD4 TEM_4 to 0.612 for CD4 TCM_3. Nine of 12 marker-enrichment tests retained BH-FDR $q < 0.05$; CD4 TEM_3 ADT, CD4 TEM_4 ADT and CD8 Naive RNA did not. Because donor-pair summaries overlap and multiple marker sets were inspected, these values are descriptive marker-concordance checks rather than family-wise confirmatory discoveries.

Seed-pair mean Spearman values were 0.644–0.843 for RNA and 0.519–0.758 for ADT, generally higher than donor-pair stability. Integrated gradients used seed 42, up to eight correct cells per donor/class and 16 integration steps. Gradient-times-input versus integrated-gradient Spearman was usually above 0.70, but CD4 TEM_4 sometimes had only one or two correct samples. The sanity analysis is therefore directional, not an independent interpretation benchmark.

12 Secondary breadth evidence

Earlier random-cell analyses are retained for completeness but are not part of the primary nested-donor statistical claim. The fixed 100,000-cell PBMC random split placed every donor in training, validation and test. Ten learned-HSR seeds reached 0.9209 ACC, 0.9200 W-F1 and 0.8437 M-F1, but paired HSR effects crossed zero. This protocol measures within-cohort architecture performance and must not be called unseen-donor generalization.

The E-MTAB-10026 patient-disjoint split assigned 90/20/20 nonoverlapping patients and 453,073/81,318/112,975 cells to training/validation/test. MOSAIC versus an early-fusion MLP reached 0.8590/0.8550/0.6471 versus 0.8553/0.8514/0.6411 for ACC/W-F1/M-F1 under this separate patient-disjoint protocol. ACC and W-F1 paired intervals were positive; the three-seed M-F1 interval crossed zero. The single-patient B_malignant label was train-only, and these values are not pooled with the primary PBMC donor inference.

GSE164378 5P used the same eight donors as the primary 3P source and a smaller 54-protein panel. A fixed three-seed MOSAIC majority ensemble reached 0.8646/0.8617/0.6916 for ACC/W-F1/M-F1. It is assay-cohort and ensemble evidence, not independent-donor or single-model superiority; the aggregation label is preserved in Table S23.

COMBAT/COVID parent-safe experiments compared an attributable MOSAIC DCRG variant, scNym and scANVI on aligned seeds 44/45/46. The attributable MOSAIC DCRG row reached 0.9651 parent-safe ACC, 0.9732 W-F1 and 0.7822 M-F1. The separate cross-method consensus, which consumes predictions from all three methods and is not a MOSAIC single-model result, reached 0.9788 ACC, 0.9790 W-F1 and 0.8372 M-F1; it is excluded from Table S23. COMBAT source provenance is recorded in `docs/datasets/combat_citeseq/README.md`.

13 Macro precision and recall decomposition

The primary M-F1 and balanced-accuracy gains over the matched MLP reflect both macro precision and macro recall, with the precision component larger in magnitude. Per fold–seed, macro precision, macro recall and macro F1 were computed from supported classes in the saved `per_class_metrics.csv` files; seeds were then averaged within each donor before paired statistics.

Table S22: Support-aware macro precision, recall and F1 decomposition for MOSAIC full versus the matched MLP protocol. Differences are MOSAIC minus MLP after averaging seeds within each held-out donor. Macro recall is the support-aware balanced-accuracy calculation. The primary M-F1 in Table S2 is the run-level aggregate, whereas this table recomputes macro F1 from supported per-class files before donor pairing; the small numerical difference is therefore an aggregation/zero-support convention difference, not a second primary endpoint. Relative to the primary run-level M-F1, the support-aware macro-F1 differs by +0.0018 for MOSAIC and +0.0017 for the MLP.

Metric	MLP matched protocol	MOSAIC full	Paired difference
Macro precision	0.8145 [0.8005, 0.8285]	0.8316 [0.8194, 0.8437]	+0.0171 [0.0111, 0.0230], $P = 0.000248$
Macro recall	0.8011 [0.7821, 0.8201]	0.8085 [0.7890, 0.8280]	+0.0074 [0.0013, 0.0135], $P = 0.0239$
Macro F1	0.7978 [0.7792, 0.8165]	0.8090 [0.7928, 0.8252]	+0.0111 [0.0062, 0.0160], $P = 0.00105$

14 Protocol-stratified breadth matrix

The breadth matrix below is a supplementary descriptive export of the existing cross-dataset artifacts. It contains no COMBAT/COVID-shift consensus row: the previously reported COMBAT consensus consumed predictions from MOSAIC, scNym and scANVI and therefore cannot be attributed to a single MOSAIC model. The GSE164378 5P MOSAIC row is a fixed three-seed majority ensemble; the PBMC and COVID blocks use their source-specific seed aggregation. Because split units, label spaces, preprocessing and aggregation modes differ, cells are shown without a pooled best-method ranking.

Table S23: Protocol-stratified secondary breadth matrix. Methods are rows and dataset–metric pairs are grouped by dataset. Each block reports ACC, W-F1, M-F1, aggregation and n seeds/artifacts; "single model" denotes one prediction artifact, "seed mean" an arithmetic mean across the listed model seeds, and "majority ensemble" a fixed vote across the listed prediction files. The PBMC strict L3, COVID-19 full and GSE164378 exact-L3 blocks retain their source protocols. MOSAIC GSE164378 is a fixed three-seed majority ensemble; the table is descriptive and is not evidence for a universal ranking. All 63 metric cells and their aggregation metadata are covered by source metadata in `results/tables/v30_main_wide_performance_table_ci_sources_2026-07-23.csv`.

Method	PBMC strict L3					COVID-19 full					GSE164378 exact-L3				
	ACC	W-F1	M-F1	Aggregation	n	ACC	W-F1	M-F1	Aggregation	n	ACC	W-F1	M-F1	Aggregation	n
MOSAIC	0.9209	0.9200	0.8437	seed mean	10	0.8656	0.8620	0.6655	seed mean	3	0.8646	0.8617	0.6916	majority ensemble	3
MLP / early fusion	0.9123	0.9105	0.8129	single model	1	0.8597	0.8552	0.6551	seed mean	3	0.8600	0.8571	0.6513	majority ensemble	3
RNA logistic	0.8779	0.8769	0.7529	single model	1	0.8133	0.8188	0.6688	seed mean	3	0.8311	0.8318	0.6110	seed mean	3
CellTypist	0.8724	0.8661	0.6930	single model	1	0.8215	0.8123	0.5600	single model	1	0.8273	0.8206	0.5780	single model	1
scANVI	0.8907	0.8882	0.7729	single model	1	0.8313	0.8222	0.5325	single model	1	0.8573	0.8530	0.6287	single model	1
totalVI	0.8100	0.8251	0.6916	single model	1	0.7169	0.7285	0.4500	single model	1	0.7853	0.7939	0.5702	single model	1
scNym	0.8570	0.8504	0.6543	single model	1	0.8426	0.8368	0.6219	single model	1	0.8443	0.8369	0.6096	single model	1

15 Panel-aware training sensitivity

The training-only panel-aware intervention is reported after the fixed breadth matrix as Supplementary Table S23. Each row averages three seeds within each held-out donor and reports two-sided 95% Student-t intervals across eight donor means. The panel-aware model uses independent 30% ADT feature dropout during training only; all evaluation masks are deterministic and selected without test labels.

Table S24: Training-only panel-aware sensitivity. The panel-aware model uses independent 30% ADT feature dropout during training only. Frozen-checkpoint rows are the reference from the deterministic mask audit; evaluation scenarios are applied after training.

Training	Evaluation	ACC	W-F1	M-F1	ECE	n donors
Frozen checkpoint	full	0.9119 [0.9086,0.9153]	0.9108 [0.9062,0.9153]	0.8072 [0.7917,0.8227]	0.0713 [0.0645,0.0781]	8
Panel-aware training (ADT dropout 0.30)	full	0.9098 [0.9058,0.9138]	0.9084 [0.9032,0.9136]	0.8016 [0.7874,0.8158]	0.0659 [0.0579,0.0739]	8
Frozen checkpoint	random_30	0.9035 [0.8988,0.9082]	0.9020 [0.8963,0.9076]	0.7904 [0.7742,0.8066]	0.0755 [0.0682,0.0827]	8
Panel-aware training (ADT dropout 0.30)	random_30	0.9049 [0.9002,0.9097]	0.9034 [0.8976,0.9091]	0.7914 [0.7766,0.8063]	0.0708 [0.0615,0.0801]	8
Frozen checkpoint	marker_tcell	0.9029 [0.8986,0.9072]	0.9021 [0.8969,0.9072]	0.7859 [0.7709,0.8008]	0.0748 [0.0664,0.0832]	8
Panel-aware training (ADT dropout 0.30)	marker_tcell	0.9043 [0.9000,0.9087]	0.9032 [0.8978,0.9086]	0.7884 [0.7748,0.8020]	0.0700 [0.0610,0.0790]	8
Frozen checkpoint	rna_only	0.8893 [0.8846,0.8940]	0.8882 [0.8826,0.8938]	0.7558 [0.7390,0.7726]	0.1063 [0.0914,0.1212]	8
Panel-aware training (ADT dropout 0.30)	rna_only	0.8898 [0.8832,0.8965]	0.8889 [0.8816,0.8963]	0.7558 [0.7363,0.7754]	0.1038 [0.0927,0.1150]	8

16 RNA library-size normalization sensitivity

We repeated the eight-fold, three-seed donor-disjoint matrix after applying CP10K using each cell's full-transcriptome RNA denominator, followed by log1p, before the train-only 3,000-gene selection. This sensitivity isolates library-size normalization from the locked source-count log1p path; ADT preprocessing, donor mapping, seeds, losses and checkpoint selection were unchanged. The CP10K path is therefore reported as a preprocessing sensitivity rather than a replacement for the prespecified primary protocol.

Table S25: Full-transcriptome RNA library-size normalization sensitivity. Values are donor means with two-sided 95% Student-t intervals after averaging three seeds within each held-out donor. The CP10K denominator uses all 20,729 stored RNA features before train-only gene selection; no validation or test labels selected the transform.

Method	RNA transform	ACC	W-F1	M-F1	n donors
Matched MLP	locked log1p	0.9103 [0.9065, 0.9141]	0.9092 [0.9046, 0.9137]	0.7961 [0.7786, 0.8136]	8
Matched MLP	full-transcriptome CP10K + log1p	0.9094 [0.9050, 0.9137]	0.9083 [0.9032, 0.9134]	0.7949 [0.7769, 0.8128]	8
MOSAIC full	locked log1p	0.9119 [0.9086, 0.9153]	0.9108 [0.9062, 0.9153]	0.8072 [0.7917, 0.8227]	8
MOSAIC full	full-transcriptome CP10K + log1p	0.9122 [0.9088, 0.9156]	0.9107 [0.9063, 0.9152]	0.8088 [0.7928, 0.8248]	8
MOSAIC–MLP	locked log1p	0.0016 [-0.0011, 0.0044]	0.0016 [-0.0009, 0.0041]	0.0111 [0.0062, 0.0160]	8
MOSAIC–MLP	CP10K + log1p	0.0029 [0.0002, 0.0056]	0.0024 [-0.0001, 0.0050]	0.0139 [0.0059, 0.0220]	8

17 Reproducibility and artifact contract

The formal matrices contain 96 primary PBMC retraining units (including the matched-protocol MLP), 24 additional parameter-matched capacity-control units, 96 branch-sensitivity retraining units, 24 full-checkpoint ablation units, 192 panel conditions, 24 panel-aware training units with training-only ADT feature dropout, 15 leave-class-out training units, 90 reject operating conditions, 72 full-training classical/tree-baseline donor runs, eight full-training CellTypist-compatible donor folds, one fixed-split scANVI run, one fixed-split totalVI run, one donor-paired MOSAIC-versus-XGBoost audit, 48 CLR sensitivity units (24 fold–seed units per method), 48 full-transcriptome CP10K sensitivity units (24 fold–seed units per method) and 24 validation-only temperature-scaling units. PDC101 contains three MOSAIC seeds and one official MMoCHi workflow artifact. The secondary breadth matrix in Table S23 contains 63 method–dataset–metric cells and is covered by the source-audit CSV cited in its caption; each cell retains source paths and the reported MOSAIC aggregation variant. Versioned script names (`v30`, `v33`, `v36`, `v50`, `v7`) denote artifact-generation snapshots rather than manuscript versions; the manuscript reports the current evidence registry and the user-owned Figure 1 asset separately. Each serious training run saves configuration, log, checkpoint, prediction, per-class metrics and artifact index.

Primary table and figure scripts read only complete declared sources:

- `experiments/exp_baseline_matrix/build_v50_matched_mlp_paired.py`;
- `experiments/exp_baseline_matrix/build_v50_macro_precision_recall.py`;
- `experiments/exp_baseline_matrix/build_v50_preprocessing_audit.py`;
- `experiments/exp_baseline_matrix/plot_v36_evidence_figure.py`;
- `experiments/exp_explainability/mosaic_n_v30/plot_v30_fig1_architecture.py` (historical generator; Figure 1 is the user-owned final asset);
- `results/tables/mosaic_n_v50_matched_mlp_paired_2026-08-03.csv`;
- `results/tables/mosaic_n_v50_hsr_scope_summary_2026-08-03.csv`;
- `experiments/exp_baseline_matrix/run_v45_pbmc_celltypist_native.py`;
- `experiments/exp_baseline_matrix/build_v7_mlp_large_audit.py`;
- `experiments/exp_explainability/mosaic_n_v50/build_v7_temperature_scaling_audit.py`;

- `experiments/exp_explainability/mosaic_n_v50/build_v7_audit_throughput.py`;
- `experiments/exp_generalization/mosaic_n_v33/run_v7_panel_aware_matrix.py`;
- `experiments/exp_missing_modality/mosaic_n_v33/evaluate_v33_panel_robustness.py`;
- `experiments/exp_generalization/mosaic_n_v33/build_v7_panel_aware_summary.py` and `build_v7_panel_aware_table.py`;
- `experiments/exp_generalization/mosaic_n_v33/build_pbmc_nested_donor_cache.py` with the CP10K sensitivity path;
- `experiments/exp_generalization/mosaic_n_v33/run_v7_rna_cp10k_matrix.py` and `build_v7_rna_cp10k_summary.py`;
- `results/tables/mosaic_n_v7_pbmc_source_provenance_2026-08-11.csv`.

The release bundle stages code, configuration, one P1/seed-42 checkpoint, ten transformed demonstration cells, compact evidence tables, commands, environment metadata and SHA-256 hashes. It excludes raw participant data and full caches. The corresponding public GitHub snapshot is tagged `v7.0.3-submission` under the MIT License and includes the checkpoint, CPU demo, compact evidence tables and checksum manifest; no Zenodo DOI, raw-data archive or full prediction archive is claimed. The panel-aware and RNA-CP10K sensitivity artifacts are included as compact evidence records rather than as raw-data redistribution.

18 Claim boundaries

Taken together, the package establishes an evidence-emitting annotation workflow with donor-disjoint M-F1 and balanced-accuracy gains over a loss- and selection-matched MLP, quantitative error auditing, measurable but panel-dependent degradation under missing proteins, target-dependent rejection, a compatible PDC101 hierarchy comparison and moderate cross-donor attribution stability. The scope is deliberately protocol-defined: ACC and W-F1 superiority is not established by the matched-MLP intervals; the near-capacity control narrows but does not remove the M-F1 gap, so the result supports a joint representation-and-evidence advantage rather than strict single-module causality; HSR, KD and learned gating are retained for evidence emission and local audit rather than claimed as aggregate closed-set drivers; and missing-panel and leave-class-out results characterize specific stresses rather than arbitrary missing modalities or unrestricted open-set recognition. Attribution remains functional and descriptive rather than causal, and the breadth matrix is protocol-stratified rather than a universal ranking across datasets.